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China-Linked SprySOCKS Backdoor Expands to Windows with Driver-Based Stealth

Malware Analysis Report

Classification	TLP:CLEAR
Published	2026-06-17
Version	1.0
Author	Lost Boys Cyber
Report Type	Malware Analysis Report

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China-Linked SprySOCKS Backdoor Expands to Windows with Driver-Based Stealth

1. Executive Summary

ESET Research disclosed two previously undocumented Windows variants of SprySOCKS, a backdoor previously reported as a Linux-focused FishMonger tool. The Windows variants are internally labeled WIN_DRV and WIN_PLUS. Both implement hard-coded command-and-control configuration, support TCP, UDP, and WebSocket communication, and expose more than 30 operator commands for host discovery, process and service control, file management, SOCKS proxying, interactive command execution, and file transfer.

The most consequential finding is the WIN_DRV variant's driver-backed stealth layer. ESET reports that WIN_DRV deploys a loader, encrypted payload containers, a driver loader, and a RawWNPF kernel driver to hide files, processes, registry keys, and network connections. The driver also enables passive TCP traffic diversion: specially crafted inbound TCP data sent to any open port can be redirected to the hidden backdoor listener, allowing operators to reach the implant without exposing the true listening port in normal host tooling.

Lost Boys Cyber assesses this activity as an intrusion lead, not a simple malware-event-only alert. The available reporting ties the Windows SprySOCKS variants to FishMonger, also tracked as Earth Lusca, TAG-22, Aquatic Panda, and Red Dev 10, and describes government victimology in Honduras, Taiwan, Thailand, and Pakistan between 2023 and 2024. Organizations that operate exposed Windows servers, print services, or administrative hosts should prioritize host scoping, driver and print-processor review, network egress analysis, and follow-on investigation for credential theft or broader espionage activity.

This report is source-backed rather than sample-backed. Lost Boys Cyber did not receive the malware samples directly and did not execute the binaries in a local sandbox. Technical details below are therefore derived from ESET's primary technical analysis and The Hacker News secondary reporting, with defensive detections and hunting logic authored by Lost Boys Cyber for validation in client telemetry.

Key Finding	Defensive Significance	Confidence
SprySOCKS now has Windows variants, WIN_DRV and WIN_PLUS	Expands FishMonger capability from Linux-focused tradecraft into Windows estate compromise	High
WIN_DRV uses a RawWNPF kernel driver	Standard user-mode process, file, registry, and network inspection may miss active implant artifacts	High
WIN_PLUS uses print processor persistence and an encrypted container under spool paths	Windows print subsystem telemetry becomes a high-value hunt source	High
C2 supports TCP, UDP, and WebSocket with shared AES/base64 framing	Network monitoring should hunt for protocol-specific magic values and unusual multi-protocol use to the same host	Medium-High
Limited UEFI bootkit indications were reported	Firmware persistence should be considered only when host artifacts support it; do not over-scope without evidence	Low-Medium

2. Sample Metadata

Field	Value
Malware Family	SprySOCKS

Variants	WIN_DRV; WIN_PLUS
Malware Type	Windows backdoor / RAT with SOCKS proxy, keylogging, file operations, service control, and driver-backed stealth in WIN_DRV
Platform	Microsoft Windows x64; Windows server hosts are specifically relevant because ESET noted server operating systems on some victim devices
Primary Sample Hash	No SHA256 available in public reporting reviewed for this report; ESET published SHA1 indicators
Additional Hashes	SHA1 indicators are captured in Section 11 and in iocs.csv
File Type / Format	ZIP archive; batch script; DLL loaders; encrypted .dat containers; Windows kernel drivers; legitimate signed DLL side-loading host binaries
Delivery / Access Assessment	Initial access not confirmed; FishMonger commonly targets public-facing servers via misconfiguration or N-day vulnerabilities
First Observed / Reported	Activity observed by ESET in 2023-2024; public reporting 2026-06-16
Threat Actor	FishMonger, assessed as China-aligned and believed by ESET to be operated by I-SOON; aliases include Earth Lusca, TAG-22, Aquatic Panda, Red Dev 10
Primary Objective	Long-term cyberespionage access, covert remote control, file collection, proxying, and host discovery
Analyst Confidence	High for capability descriptions from ESET; medium for initial access path; low-medium for UEFI bootkit involvement

3. Source Review & Confidence

Source	Contribution	Reliability / Credibility	Confidence Notes
ESET Research, FishMonger's arsenal upgraded: SprySOCKS for Windows, 2026-06-16	Primary technical analysis, variant names, component paths, C2 behavior, IOCTLs, command IDs, IOCs, ATT&CK mapping	A / 2	Direct vendor reverse-engineering and telemetry-backed reporting; treated as primary evidence
The Hacker News, China-Linked SprySOCKS Backdoor Expands to Windows with Driver-Based Stealth, 2026-06-16	Secondary summary and intake source	B / 3	Useful for prioritization and initial routing; not used as sole technical basis where ESET detail exists
Trend Micro prior SprySOCKS Linux reporting, referenced by ESET	Lineage context for Linux SprySOCKS, Trochilus, RedLeaves, HP-Socket, C2 framing, and related infrastructure	B / 3	Used only as context because the present report focuses on ESET's Windows variant findings

Directly observed from primary reporting:

- ESET identified two Windows SprySOCKS variants named WIN_DRV and WIN_PLUS.
- Both variants use hard-coded C2 configuration and support TCP, UDP, and WebSocket channels.
- WIN_DRV uses a kernel-driver chain to hide network connections, processes, files, and registry keys.
- WIN_PLUS uses print processor persistence and an encrypted container in Windows spool paths.
- ESET published SHA1 file indicators and one C2 IP address.

Assessed but not directly validated by Lost Boys Cyber:

- FishMonger attribution is accepted at high confidence based on ESET's stated assessment and code lineage to prior SprySOCKS.
- Initial access likely involved exposed or unpatched public-facing applications, but ESET did not confirm the exact vector.
- UEFI bootkit involvement remains limited-indication reporting and should be handled as a scoping hypothesis, not a confirmed finding.

4. Initial Triage and Delivery Assessment

ESET's discovery path began with a VirusTotal ZIP archive named klelam00007.zip uploaded in April 2024. The archive contained legitimate binaries used for DLL side-loading and three suspicious encrypted .dat files. ESET analysis showed that those .dat files contained the WIN_DRV Windows SprySOCKS variant, including the encrypted backdoor, DriverLoader, and RawWNPF driver. A second variant, WIN_PLUS, was later identified in ESET telemetry.

Variant / Artifact	Delivery Role	Notes
klelam00007.zip	Discovery archive for WIN_DRV	Contained side-loading host components and encrypted payload containers
klelam00007.bat	WIN_DRV persistence script	Copies files to %SystemRoot%\Fonts and creates scheduled task ApphostRagistreationVerifier
ApphostRagistreationVerifier.exe	Side-loading host	Legitimate ThinPrint executable renamed to mimic Microsoft's AppHostRegistrationVerifier.exe
tpsvc.dll	Legitimate DLL in side-loading chain	Loads malicious tpsvcloc.dll through MFC satellite DLL search behavior
tpsvcloc.dll	SprySOCKS loader	Performs environment checks, verifies persistence, decrypts payloads, and starts injection / driver loading
X1B5206BDC1743DD.dat	WIN_DRV encrypted container	Holds SprySOCKS backdoor plus encrypted driver artifacts
KX1B5206BDC1743DD.dat	Encrypted DriverLoader	Decrypted to fsdiskbit.sys before minifilter service loading
KW1B5206BDC1743FP.dat	Encrypted RawWNPF driver	Manually mapped by DriverLoader and used for stealth
VSPMsg.dll	WIN_PLUS first-stage loader	Installed as a print processor and intended to execute under spoolsv.exe
config.dat	WIN_PLUS encrypted container	Stored under C:\Windows\System32\spool\drivers\color\ and used to load the loader/backdoor

Initial access is unresolved. ESET states FishMonger often targets public-facing servers and frequently abuses server-side N-day vulnerabilities. For triage, defenders should treat a SprySOCKS finding as evidence that a public-facing host, print subsystem, or administrative Windows server may have been used as the intrusion foothold.

5.1 File and Build Characteristics

Because no local malware sample was provided, this section summarizes published static findings. The Windows SprySOCKS chain uses ordinary Windows artifact types in a layered manner: batch scripts for setup and trace removal, legitimate signed binaries for DLL side-loading, malicious DLL loaders, AES-encrypted .dat containers, process-doppelganging payload execution, and kernel drivers for stealth.

Static Signal	Evidence	Analyst Value
AES-ECB encrypted containers	ESET reports 128-bit AES-ECB with hard-coded key uXQLESMXGaRMs6BL for payload and driver containers	Supports static scanning for encrypted container names and memory-recovery focused analysis
Driver-backed rootkit component	RawWNPf creates \Device\RawWNPf and exposes custom IOCTLs	Indicates host tools can be blinded; kernel telemetry and offline disk review become important
Legitimate side-loading hosts	ThinPrint AutoConnect printer creation service executable and tpsvc.dll participate in loading tpsvcloc.dll	Enables detection logic around suspicious execution from Fonts and unexpected ThinPrint components
Print processor loader	WIN_PLUS installs VSPMsg.dll as a print processor under Windows x64 Print Processors	Makes print processor registry keys and spool paths priority host artifacts
Process doppelganging	Loader creates svchost.exe and injects shellcode using process doppelganging	High-signal execution behavior when correlated with temp files prefixed TH and unsigned parent chain
PDB path in DriverLoader	ESET reports a Chinese-language PDB path under DriverMemoryLoadDriver\x64\Release	Supports attribution context and suggests in-development driver-loading project lineage
Dynamic API resolution / encrypted strings	ESET maps dynamic API resolution and encrypted or encoded files	Reduces raw string visibility and requires behavior-based detection

5.2 Strings, Imports, and Embedded Artifacts

Artifact	Value	Relevance
AES container key	uXQLESMXGaRMs6BL	Payload and driver-container decryption key reported by ESET
C2 message encryption key	QFTHEYjzX3RBOMgZ	AES key used after base64 decoding for C2 message body data
Backdoor DLL original name	PrcsServer.dll	Shared across analyzed variants
Backdoor export	Stop	Exported function name reported in both variants
Loader mutex	fqwhi2d1qaz2	WIN_PLUS loader mutex
Backdoor mutex	prcs-server-run	Main backdoor initialization mutex

Keylogger mutex	Global\{DCAA7ED8-521B-4EAB-BE21-65254CF59239}	Created when keylogging conditions are met
Keylogger config	%appdata%\Microsoft\Vault\lgf.dat	Keylogging activates when [config] key=1 exists
Keylogger output	%appdata%\Microsoft\Vault\lg.dat	Stores XOR-encoded clipboard, active-window, and keystroke data
IFEO persistence key	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\vds.exe\debugger	WIN_DRV loader persistence fallback
Print processor key	HKLM\SYSTEM\ControlSet001\Control\Print\Environments\Windows x64\Print Processors\VSPMsg	WIN_PLUS first-stage persistence
Driver service key	HKLM service key msdiskserver	Transient minifilter driver load path
RawWNPF device	\Device\RawWNPF	Kernel driver control surface

Security-product and sandbox-evasion checks reported by ESET include searches for snxhk.dll, SxWrapper.dll, SxIn.dll, SXIn64.dll, Sbiedll.dll, and cmdvrt32.dll. The presence of these checks supports sandbox-evasion and security-software discovery mapping.

5.3 Static Configuration Clues

Channel	Host	Port	Notes
TCP client channel	207.148.78[.]36	443	Hard-coded C2 endpoint
UDP client channel	207.148.78[.]36	53	Same host, different transport
WebSocket channel	207.148.78[.]36	80	Same host, web-compatible transport
Local TCP listener	Victim host	53781	Backdoor server port allowed through a misleadingly named firewall rule
Protocol	Marker / Framing		Detection Use
TCP	12-byte header containing CRC, magic 0xACACBCBC, and data size		Candidate PCAP hunt and network IDS feature
UDP	36-byte header with magic 0xACACBFBC at offset 0x1C		Candidate packet-content signature with environment-specific tuning
WebSocket	Masking-Key 0x1BDCCBAA		Candidate WebSocket protocol anomaly signal
Message body	CRC, WORD 0x0003, AES-ECB encrypted data, base64 encoded		Useful for protocol-aware decoding and emulation

6.1 Reverse-Engineering Method

Lost Boys Cyber did not receive the binaries and did not perform local Ghidra, radare2, capa, or sandbox validation. This code-analysis section is based on ESET's published reverse-engineering results and is structured so defenders can reproduce validation if samples are later acquired. The recommended validation path is to hash-confirm the sample, isolate the side-

loading chain, decrypt containers using published keys where lawful and safe, inspect DriverLoader and RawWNPF in Ghidra, and compare IOCTL and C2 behavior against the indicators below.

6.2 Functions and Logic of Interest

Function / Routine	Evidence	Why It Matters
WIN_DRV loader environment checks	Checks for security and sandbox DLLs before continuing	Supports evasion detection and explains inconsistent sandbox execution
Persistence verification / repair	Confirms expected files exist under %SystemRoot%\Fonts and can recreate persistence	Makes simple file deletion insufficient if chain is still partly active
Container decryptor	Uses 128-bit AES-ECB key uXQLESMXGaRMs6BL to decrypt X1B5206BDC1743DD.dat and driver artifacts	Enables recovery of payloads during forensic analysis
Process doppelgänger injection	Creates svchost.exe using a token from spoolsv.exe and injects DllToShellCode-generated shellcode	Establishes suspicious process lineage and memory-scanning target
DriverLoader	Decrypts KW1B5206BDC1743FP.dat and manually maps RawWNPF	Avoids writing the final rootkit driver plainly to disk
RawWNPF IOCTL dispatcher	Provides IOCTLs to hide connections, processes, files, persistence keys, initialize functionality, and delete driver binary	Central rootkit control surface for detection and reverse engineering
WIN_PLUS first-stage loader	Runs only under spoolsv.exe, decrypts config.dat, injects loader into svchost.exe, and registers VSPMsg print processor	High-value print-spooler hunt pivot
SprySOCKS command dispatcher	Supports more than 30 message IDs for shell, process, service, file, SOCKS, and channel management	Confirms the implant supports full remote administration and staging
Keylogger module	Activates when lgf.dat [config] key=1 exists and writes XOR-encoded lgf.dat	Indicates collection is configurable and may be dormant by default

6.3 Anti-Analysis, Evasion, and Obfuscation

SprySOCKS combines several layers of defensive friction:

- Loader checks for sandbox or security product DLLs before executing.
- Payload components are stored in AES-encrypted containers rather than plainly on disk.
- Strings and APIs are obfuscated or dynamically resolved according to ESET's ATT&CK mapping.
- WIN_PLUS first-stage behavior depends on execution by spoolsv.exe, which can prevent automated sandboxes from reaching the malicious path.
- WIN_DRV hides files, registry keys, processes, and network connections via the RawWNPF kernel driver.
- WIN_DRV can redirect specially crafted inbound TCP traffic from any open port to the hidden listener, frustrating port-based triage.
- ESET reports RawWNPF uses KdDisableDebugger to disable the kernel debugger if active.

7.1 Execution Overview

No local dynamic execution was performed by Lost Boys Cyber. The following execution behavior is sourced from ESET's published analysis and should be treated as expected behavior for lab validation.

WIN_DRV execution begins with klelam00007.bat or loader-driven repair logic placing the malware set under %SystemRoot%\Fonts. Execution proceeds through a legitimate signed side-loading chain, loads tpsvcloc.dll, decrypts the encrypted payload container, starts a new svchost.exe process using a token from spoolsv.exe, injects the SprySOCKS backdoor using process doppelganging, decrypts DriverLoader to C:\Windows\System32\drivers\fsdiskbit.sys, registers a transient minifilter service named msdiskserver, loads DriverLoader with NtLoadDriver, and then allows DriverLoader to manually map RawWNPF from the encrypted driver container.

WIN_PLUS execution centers on the print subsystem. A first-stage VSPMsg.dll print processor under spoolsv.exe decrypts C:\Windows\System32\spool\drivers\color\config.dat, injects the loader into svchost.exe, registers the print processor for persistence, copies artifacts from C:\ProgramData\Microsoft Event\PFs\, and restarts the print spooler through an affair-build.bat cleanup script. The SprySOCKS loader then decrypts and injects the backdoor into svchost.exe.

7.2 Filesystem, Registry, Service, and Task Changes

Modification Type	Artifact	Evidence	Confidence
File copy	%SystemRoot%\Fonts\tpsvc.dll, tpsvcloc.dll, ApphostRagistreationVerifier.exe, X1B5206BDC1743DD.dat, KX1B5206BDC1743DD.dat, KW1B5206BDC1743FP.dat	ESET WIN_DRV loader and batch-script analysis	High
Scheduled task	ApphostRagistreationVerifier running ApphostRagistreationVerifier.exe as NT AUTHORITY\SYSTEM at startup	ESET batch-script analysis	High
IFEO debugger	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\vds.exe\debugger	ESET loader fallback persistence analysis	High
Driver drop	C:\Windows\System32\drivers\fsdiskbit.sys	ESET DriverLoader staging analysis	High
Transient service	msdiskserver minifilter service registry key	ESET NtLoadDriver setup analysis	High
Print processor	HKLM\SYSTEM\ControlSet001\Control\Print\Environments\Windows x64\Print Processors\VSPMsg	ESET WIN_PLUS loader analysis	High
Spool file	C:\Windows\System32\spool\prtprocs\x64\VSPMsg.dll	ESET telemetry and loader analysis	High
Encrypted config container	C:\Windows\System32\spool\drivers\color\config.dat	ESET WIN_PLUS analysis	High
Keylogger config/output	%appdata%\Microsoft\Vault\lgf.dat and	ESET backdoor analysis	Medium-High

	lg.dat		
Firewall rule	Core Networking - Packet Too Big(ICMPv6 - In), allowing inbound TCP on local port 53781	ESET C2 analysis	High

7.3 Network and Command-and-Control Behavior

The backdoor supports TCP, UDP, and WebSocket channels and can act as both a client and server. WIN_PLUS used 207.148.78[.]36 over TCP/443, UDP/53, and WebSocket/80 in the public analysis. A local TCP server port of 53781 was configured, and the malware modifies Windows Firewall by deleting and recreating a misleadingly named rule, Core Networking - Packet Too Big(ICMPv6 - In), to permit inbound TCP traffic to that port.

WIN_DRV may carry an empty C2 configuration and operate as a passive backdoor. In that scenario, the implant starts a TCP server on a random local port, hides the connection through RawWNPF, and relies on the driver to divert specially crafted TCP data received on any open port to the hidden backdoor server. This mode is operationally important because network defenders may see traffic to an apparently legitimate open service port while host tools fail to show the true backdoor listener.

7.4 Observed Capability Summary

Capability	Supporting Dynamic / Code Evidence
System information collection	Message ID 0x09 collects computer name, OS, network adapters, memory, CPU, privileges, language, time, and backdoor version
Interactive command shell	Message IDs 0x0A, 0x0B, and 0x0D start, write to, and stop an interactive console
Process control	Message IDs 0x11-0x13 enumerate modules/processes and terminate a process
Service control	Message IDs 0x1E-0x22 enumerate, configure, start, control, and delete services
SOCKS proxy	Message IDs 0x23-0x26 initialize and manage SOCKS proxy behavior
File transfer and operations	Message IDs 0x2A-0x2D and 0x3C-0x43 support upload/download, disk-space discovery, listing, deletion, directory creation, rename, execute, copy, and recent-file listing
Keylogging and clipboard capture	Conditional keylogger records clipboard data, active window titles, and keystrokes into XOR-encoded lg.dat
Network stealth	RawWNPF hides configured connections and diverts specially crafted inbound TCP packets
Host artifact stealth	RawWNPF hides configured files, processes, and the IFEO persistence key

8. Process Tree and Execution Chain

Step	Parent Process	Child / Action	Why It Matters
1	Operator / staging context	klelam00007.bat copies files to %SystemRoot%\Fonts and creates scheduled task	Establishes WIN_DRV persistence and unusual

		ApphostRagistreationVerifier	execution from Fonts
2	Task Scheduler / system startup	ApphostRagistreationVerifier.exe executes from %SystemRoot%\Fonts	Legitimate renamed ThinPrint executable provides side-loading cover
3	ApphostRagistreationVerifier.exe	Loads tpsvc.dll, which loads tpsvcloc.dll	DLL side-loading route into malicious loader
4	tpsvcloc.dll	Decrypts X1B5206BDC1743DD.dat and extracts driver containers	Reveals encrypted payload staging
5	tpsvcloc.dll	Creates svchost.exe using token from spoolsv.exe and injects backdoor shellcode via process doppelganging	Suspicious service-host process with print-spooler token relationship
6	tpsvcloc.dll	Drops fsdiskbit.sys and creates msdiskserver minifilter service	Kernel execution preparation
7	DriverLoader	Manually maps RawWNPF from KW1B5206BDC1743FP.dat	Activates rootkit-style stealth layer
8	RawWNPF / SprySOCKS	Hides backdoor process, files, registry, network connections, and supports traffic diversion	Complicates live triage and justifies offline evidence collection
9	spoolsv.exe	Loads VSPMsg.dll print processor in WIN_PLUS chain	Print processor persistence route for second variant
10	VSPMsg.dll / loader	Decrypts config.dat and injects SprySOCKS into svchost.exe	WIN_PLUS execution and C2 activation

Analyst note: svchost.exe alone is not a useful detection. The stronger signal is svchost.exe creation or injection in proximity to unusual Fonts, spool, print-processor, IFEO, transient-driver, and TH-prefixed temporary-file artifacts.

9. Persistence and System Modification

Modification Type	Artifact	Defensive Interpretation	Collection Priority
Scheduled task	ApphostRagistreationVerifier	High-signal WIN_DRV persistence; spelling deviates from legitimate AppHostRegistrationVerifier	Export task XML and associated action path
DLL side-loading	%SystemRoot%\Fonts\ApphostRagistreationVerifier.exe, tpsvc.dll, tpsvcloc.dll	Legitimate signed binary plus malicious DLL in abnormal directory	Hash files, preserve MFT/USN, inspect signer and load events
IFEO debugger	vds.exe debugger value	Persistence fallback and service-triggered execution path	Export registry key and transaction logs
Print processor	Print Processors\VSPMsg with Driver=VSPMsg.dll	WIN_PLUS persistence under spoolsv.exe	Export print processor keys and spool DLLs
Encrypted spool container	C:\Windows\System32\spool\driv	Backdoor/loader storage for	Preserve file, timestamps,

	ers\color\config.dat	WIN_PLUS	alternate data streams if present
Driver load	fsdiskbit.sys / msdiskserver	Transient kernel driver staging	Review Code Integrity, Sysmon driver-load, and service-control telemetry
Hidden driver device	\Device\RawWNPF	Active rootkit control device	Prefer memory acquisition and trusted offline analysis
Firewall modification	Misleading Core Networking - Packet Too Big(ICMPv6 - In) inbound TCP allow rule	Conceals local listener allowance under legitimate-looking rule name	Export firewall rules and correlate creation time to implant activity
Keylogger files	%appdata%\Microsoft\Vault\lgf.dat and lgf.dat	Indicates optional collection module activation	Preserve file content and user context

Remediation must account for rootkit behavior. On a suspected WIN_DRV host, do not rely solely on live netstat, dir, tasklist, reg query, or standard EDR console visibility. Use memory capture, offline disk acquisition, trusted recovery media, and known-good telemetry from before the suspected driver load where possible.

10. Network and Infrastructure Analysis

IOC / Infrastructure	Type	Role	Stability / Confidence
207.148.78[.]36	IP address	Hard-coded WIN_PLUS C2 over TCP/443, UDP/53, WebSocket/80	High confidence as reported IOC; medium operational stability because single IP cloud infrastructure can churn
207.148.75[.]122	IP address	Prior SprySOCKS delivery server reported by Trend Micro and referenced by ESET	Medium confidence contextual infrastructure; do not treat as exclusive to this campaign without corroboration
207.148.64.0/20	Hosting range	Vultr range containing reported C2 and prior delivery server	Low as a block indicator; useful for enrichment and scope review, not broad blocking without additional evidence
TCP magic 0xACACBCBC	Protocol marker	TCP C2 framing	Medium-High for packet capture matching with protocol constraints
UDP magic 0xACACBFBC	Protocol marker	UDP C2 framing at offset 0x1C in 36-byte header	Medium-High for packet capture matching with offset validation
WebSocket masking key 0x1BDCCBAA	Protocol marker	WebSocket C2 masking key	Medium for anomaly hunting; requires tuning to avoid protocol-parser false positives
Local TCP port 53781	Host listener	Configured WIN_PLUS backdoor local server port	Medium; can vary by configuration and may be hidden by RawWNPF

Network defensive implications:

- Multi-protocol communication to the same IP over TCP/443, UDP/53, and HTTP/WebSocket-compatible TCP/80 should be investigated when tied to suspicious Windows server hosts.
- UDP/53 traffic to 207.148.78[.]36 should not be assumed to be DNS without payload inspection.
- Host-based network enumeration can be unreliable if RawWNPF is active; compare endpoint telemetry with span/tap, firewall, proxy, EDR network sensors, and NetFlow.
- Traffic-signaling behavior means an attacker may reach the implant by sending crafted data to an otherwise expected open service port.

11. Indicators of Compromise

The following indicators are public-source indicators from ESET. They should be used for scoping and enrichment, not as the only detection layer. The strongest operational approach is to correlate these IOCs with suspicious persistence, print subsystem changes, hidden driver behavior, and C2 framing.

IOC Type	Value / Count	Source	Operational Notes
SHA1	11 published file hashes	ESET	Includes encrypted containers, loaders, drivers, archive, and batch script
IP address	207.148.78[.]36	ESET	Hard-coded WIN_PLUS C2
Contextual IP address	207.148.75[.]122	ESET citing Trend Micro	Prior SprySOCKS delivery server in same Vultr /20
File paths	Fonts, spool, drivers, Vault paths listed below	ESET	Strong host-scoping pivots when timestamps align
Registry paths	IFEO vds.exe debugger; Print Processors\VSPMsg	ESET	High-value persistence checks
Mutexes	fqwhi2d1qaz2; prcs-server-run; Global\{DCAA7ED8-521B-4EAB-BE21-65254CF59239}	ESET	Useful for memory/live-response validation when not hidden
SHA1	Filename	Description	
955BFC3DCC867256F9F46A606DEB0779FA3416D8	KX1B5206BDC1743DD.dat	Encrypted SprySOCKS DriverLoader driver	
44DC4A08C5EB0972C8E18B0E01284E06F09006BB	bthcam.sys	SprySOCKS DriverLoader driver	
AB87B29B6F79487C75CA08D102E79001E536F083	KW1B5206BDC1743FP.dat	Encrypted SprySOCKS RawWNPF driver	
6490B8E4AADE25A3EE2DA9A47F312DB2122470BC	X1B5206BDC1743DD.dat	Encrypted container for WIN_DRV backdoor, RawWNPF, and DriverLoader	
E7484C24B88A1A2407A8F09D734F9A993670285B	klelam00007.zip	ZIP archive containing WIN_DRV and components	
621D1952839BE4B0A1B0E66E87BCE5062CA368ED	tpsvcloc.dll	SprySOCKS loader	

2457EED2AB28E37741F10914EF929DA D2C8079D4	VSPMsg.dll	First-stage loader for WIN_PLUS
D2C706B1EAF662BF0CE124B5032F73E D84BDA24A	N/A	WIN_PLUS SprySOCKS backdoor
5F3B87CEF56683D9A9E19186E0FD0D8 019B559C4	N/A	SprySOCKS loader
C793CA31E3F6628B5C8986146953BF66 232E9A30	config.dat	Encrypted WIN_PLUS container
037DB2445F3D72388CB2CF8510563148 E5A184BE	N/A	Batch script that persists WIN_DRV

12. MITRE ATT&CK Mapping

Technique	Name	Evidence
T1592.004	Gather Victim Host Information: Client Configurations	SprySOCKS can collect OS, CPU, memory, privileges, language, and version details
T1590.005	Gather Victim Network Information: IP Addresses	System information collection includes network adapter and IP information
T1587.001	Develop Capabilities: Malware	FishMonger developed Windows SprySOCKS variants and driver components
T1059.003	Command and Scripting Interpreter: Windows Command Shell	Interactive console commands supported by message IDs 0x0A-0x0D
T1053.005	Scheduled Task/Job: Scheduled Task	WIN_DRV scheduled task ApphostRagistreationVerifier starts at boot
T1569.002	System Services: Service Execution	Service-based execution and driver loading via msdiskserver and vds.exe path
T1106	Native API	Uses CreateProcessAsUserW, NtLoadDriver, WFP APIs, and other Windows native APIs
T1547.012	Boot or Logon Autostart Execution: Print Processors	WIN_PLUS installs VSPMsg print processor
T1546.012	Event Triggered Execution: Image File Execution Options Injection	WIN_DRV writes vds.exe debugger persistence
T1205.002	Traffic Signaling: Socket Filters	RawWNPF diverts specially crafted TCP packets to a hidden backdoor listener
T1134.002	Access Token Manipulation: Create Process with Token	Loader creates svchost.exe using a token obtained from spoolsv.exe
T1622	Debugger Evasion	RawWNPF reportedly invokes KdDisableDebugger

T1140	Deobfuscate/Decode Files or Information	Loaders decrypt AES-encrypted containers and strings
T1070.004	Indicator Removal: File Deletion	Cleanup scripts remove original deployment files
T1070.009	Indicator Removal: Clear Persistence	Loader removes transient driver-service registry artifacts after loading
T1027.007	Obfuscated Files or Information: Dynamic API Resolution	ESET reports dynamic API resolution
T1027.013	Obfuscated Files or Information: Encrypted/Encoded File	Payloads and drivers stored in encrypted .dat containers
T1055.013	Process Injection: Process Doppelgänger	Loaders inject into svchost.exe through process doppelgänger
T1014	Rootkit	RawWNPF hides processes, files, registry keys, and network connections
T1497	Virtualization/Sandbox Evasion	Loader checks security/sandbox DLLs and WIN_PLUS requires spoolsv.exe execution
T1574.002	Hijack Execution Flow: DLL Side-Loading	Legitimate ThinPrint components load malicious tpsvcloc.dll
T1562.004	Disable or Modify System Firewall	Malware modifies a firewall rule to allow inbound TCP to local listener
T1010	Application Window Discovery	Keylogger captures active foreground window title
T1083	File and Directory Discovery	File-listing command support and recent-file enumeration
T1518.001	Software Discovery: Security Software Discovery	Checks for security and sandbox DLLs
T1082	System Information Discovery	Host system information command support
T1614.001	System Location Discovery: System Language Discovery	Collects system language
T1007	System Service Discovery	Enumerates services
T1124	System Time Discovery	Collects current time
T1056.001	Input Capture: Keylogging	Optional keylogger writes lg.dat
T1115	Clipboard Data	Keylogger records clipboard content
T1132.001	Data Encoding: Standard Encoding	C2 body is base64 encoded before AES processing
T1573.001	Encrypted Channel: Symmetric Cryptography	C2 data uses AES
T1008	Fallback Channels	TCP, UDP, and WebSocket channels

		supported
T1665	Hide Infrastructure	RawWNPf hides active network connections
T1571	Non-Standard Port	Backdoor listener may use nonstandard local ports such as 53781
T1095	Non-Application Layer Protocol	Custom TCP/UDP protocol support
T1041	Exfiltration Over C2 Channel	File upload capability over C2

13.1 Detection Logic Summary

Signal	Telemetry Source	Analytic Goal	Tuning Notes
ApphostRagistreationVerifier scheduled task	Windows Security 4698, Sysmon Event ID 1/12/13, TaskScheduler Operational	Detect WIN_DRV boot persistence	Note misspelling Ragistreation; compare with legitimate AppHostRegistrationVerifier
Execution from %SystemRoot%\Fonts	Sysmon process/file events, EDR process telemetry	Detect abnormal side-loading and payload staging	Fonts directory may contain font files, not executable chains
IFEO debugger for vds.exe	Sysmon registry events, Windows registry audit	Detect fallback persistence	Very rare in normal enterprise environments
VSPMsg print processor	Registry events, spooler operational logs, EDR module loads	Detect WIN_PLUS persistence	Baseline legitimate print processors per server role
Driver load fsdiskbit.sys or msdiskserver	Sysmon Event ID 6, CodeIntegrity, System 7045	Detect driver-backed stealth activation	Rootkit may delete files/keys after load; correlate short-lived artifacts
Firewall rule rename / recreate	Windows Firewall event logs, PowerShell/command-line telemetry	Detect listener exposure	Rule name appears legitimate but allows inbound TCP to backdoor port
Multi-protocol C2 to 207.148.78.36	Firewall/proxy/NetFlow/EDR network telemetry	Detect reported WIN_PLUS C2	Treat as IOC hit and scope around host artifacts
C2 magic values	Full packet capture, IDS, Zeek custom parser	Detect SprySOCKS protocol framing	Validate byte order and offsets before production blocking
Keylogger Vault artifacts	File events, EDR file telemetry	Detect optional collection module	Avoid broad alerting on Microsoft\Vault alone without lgf.dat/lg.dat specificity

13.2 Detection Content

```

title: Potential SprySOCKS WIN_DRV Scheduled Task Persistence
id: 4a1c9a2e-8cb2-4b42-9e64-000000000024
status: experimental
description: Detects creation or execution of the misspelled ApphostRagistreationVerifier scheduled task reported in Windows SprySOCKS WIN_DRV activity.
references:
  - https://www.welivesecurity.com/en/eset-research/fishmongers-arsenal-upgraded-sprysocks-windows/
author: Lost Boys Cyber
date: '2026-06-17'
tags:

```



```

- attack.persistence
- attack.t1053.005
- malware.sprysocks
logsource:
  product: windows
detection:
  selection_task:
    EventID:
      - 4698
      - 4702
    TaskName|contains: 'ApphostRagistreationVerifier'
  selection_proc:
    Image|endswith: '\\ApphostRagistreationVerifier.exe'
  selection_cmd:
    CommandLine|contains: 'ApphostRagistreationVerifier'
  condition: 1 of selection_*
falsepositives:
  - Expected to be rare; verify spelling and path before suppressing.
level: high

title: Potential SprySOCKS WIN_PLUS Print Processor Persistence
id: de63022b-0a37-4f96-a66d-000000000024
status: experimental
description: Detects VSPMsg print processor persistence and related spool path artifacts associated with SprySOCKS WIN_PLUS.
references:
  - https://www.welivesecurity.com/en/eset-research/fishmongers-arsenal-upgraded-sprysocks-windows/
author: Lost Boys Cyber
date: '2026-06-17'
tags:
  - attack.persistence
  - attack.t1547.012
  - malware.sprysocks
logsource:
  product: windows
detection:
  selection_registry:
    TargetObject|contains|all:
      - '\\Control\\Print\\Environments\\Windows x64\\Print Processors\\VSPMsg'
  selection_file:
    TargetFilename|endswith:
      - '\\System32\\spool\\prtprocs\\x64\\VSPMsg.dll'
      - '\\System32\\spool\\drivers\\color\\config.dat'
  selection_process:
    Image|endswith: '\\spoolsv.exe'
    ImageLoaded|endswith: '\\VSPMsg.dll'
  condition: 1 of selection_*
falsepositives:
  - Custom print processor software named VSPMsg; validate file signature, hash, and source path.
level: high

rule MAL_WIN_SprySOCKS_Windows_Public_Artifacts_Jun2026 {
  meta:
    description = "Detects public artifact strings and constants associated with Windows SprySOCKS reporting"
    author = "Lost Boys Cyber"
    date = "2026-06-17"
    source = "ESET public reporting"
    confidence = "medium"
    tlp = "CLEAR"
  strings:
    $s1 = "prcs-server-run" ascii wide
    $s2 = "fqwhi2d1qaz2" ascii wide
    $s3 = "Global\\{DCAA7ED8-521B-4EAB-BE21-65254CF59239}" ascii wide
    $s4 = "uXQLESMXGaRMs6BL" ascii wide
    $s5 = "QFTHEYjzX3RBOMgZ" ascii wide
    $s6 = "\\Device\\RawWPNPF" ascii wide

```

```

$S7 = "ApphostRagistreationVerifier" ascii wide
$S8 = "VSPMsg.dll" ascii wide
$S9 = "X1B5206BDC1743DD.dat" ascii wide
$S10 = "KW1B5206BDC1743FP.dat" ascii wide
$m_tcp = { BC BC AC AC }
$m_udp = { BC BF AC AC }
$m_ws = { AA CB DC 1B }
condition:
    uint16(0) == 0x5A4D and (2 of ($S*) or (1 of ($S*) and 1 of ($m_*)))
}

// Microsoft Defender XDR: suspicious SprySOCKS persistence and side-loading pivots
let lookback = 30d;
let suspiciousNames = dynamic(["ApphostRagistreationVerifier", "tpsvclloc.dll", "VSPMsg.dll",
"X1B5206BDC1743DD.dat", "KX1B5206BDC1743DD.dat", "KW1B5206BDC1743FP.dat", "config.dat",
"fsdiskbit.sys", "bthcam.sys"]);
DeviceFileEvents
| where Timestamp > ago(lookback)
| where FolderPath has_any (suspiciousNames)
    or FolderPath has @"\Windows\Fonts\tpsvclloc.dll"
    or FolderPath has @"\Windows\System32\spool\prtprocs\x64\VSPMsg.dll"
    or FolderPath has @"\Windows\System32\spool\drivers\color\config.dat"
    or FolderPath has @"\Windows\System32\drivers\fsdiskbit.sys"
| project Timestamp, DeviceName, ActionType, FolderPath, SHA1, InitiatingProcessFileName,
InitiatingProcessCommandLine

// Microsoft Defender XDR: SprySOCKS network IOC and multi-protocol C2 pattern
let lookback = 30d;
let c2 = dynamic(["207.148.78.36", "207.148.75.122"]);
DeviceNetworkEvents
| where Timestamp > ago(lookback)
| where RemoteIP in (c2)
    or (RemotePort in (53, 80, 443) and RemoteIP startswith "207.148.")
| summarize Ports=make_set(RemotePort), Protocols=make_set(Protocol),
ProcessNames=make_set(InitiatingProcessFileName), FirstSeen=min(Timestamp), LastSeen=max(Timestamp),
Count=count() by DeviceName, RemoteIP
| where RemoteIP in (c2) or array_length(Ports) >= 2
index=wineventlog earliest=-30d
(
    "ApphostRagistreationVerifier" OR "VSPMsg.dll" OR "fsdiskbit.sys" OR "msdiskserver" OR
    "tpsvclloc.dll" OR "RawWNPF"
)
| table _time host source EventCode Image CommandLine TargetObject TargetFilename Details

```

13.3 Detection Validation and Blind Spots

Validation steps: test rules against historical Windows, Sysmon, EDR, and print-spooler logs; confirm whether the environment logs image loads from spoolsv.exe, driver-load events, task-creation events, and registry changes under IFEO and Print Processors; use C2 IP matching as high-priority enrichment but do not rely on IP-only blocking.

Blind spots: RawWNPF may hide active connections from local enumeration tools; WIN_DRV can run without hard-coded remote C2 and rely on passive TCP traffic diversion; file-name detections are vulnerable to operator renaming; the YARA rule is public-artifact based and should be used for triage, not sole attribution.

14.1 Hunt Hypotheses

- A compromised Windows server may contain SprySOCKS WIN_DRV artifacts under %SystemRoot%\Fonts and persistence through a misspelled ApphostRagistreationVerifier scheduled task or IFEO debugger key for vds.exe.
- A compromised Windows print server or host running the Print Spooler service may contain WIN_PLUS persistence through a VSPMsg print processor and encrypted config.dat under spool\drivers\color.
- A host communicating with 207.148.78.36 over more than one of TCP/443, UDP/53, and WebSocket/80 may be using WIN_PLUS C2 rather than normal web or DNS traffic.
- A suspected WIN_DRV host may show mismatches between external network telemetry and local process/socket visibility because RawWNPF hides active connections.

- Keylogger activation may be visible through Microsoft\Vault\lgf.dat and lgf.dat artifacts even when the main implant is otherwise hidden.

14.2 Hunt Queries

```
// Hunt 1: WIN_DRV persistence under Fonts, scheduled tasks, and IFE0 debugger keys
let lookback = 45d;
let names = dynamic(["ApphostRagistreationVerifier", "tpsvclloc.dll", "X1B5206BDC1743DD.dat",
"KX1B5206BDC1743DD.dat", "KW1B5206BDC1743FP.dat"]);
union DeviceProcessEvents, DeviceFileEvents, DeviceRegistryEvents
| where Timestamp > ago(lookback)
| where FolderPath has_any (names)
    or InitiatingProcessCommandLine has_any (names)
    or FileName in~ (names)
    or RegistryKey has @"Image File Execution Options\vds.exe"
    or RegistryValueData has "ApphostRagistreationVerifier"
| project Timestamp, DeviceName, ActionType, FileName, FolderPath, ProcessCommandLine,
InitiatingProcessFileName, InitiatingProcessCommandLine, RegistryKey, RegistryValueName,
RegistryValueData
| order by Timestamp desc

// Hunt 2: WIN_PLUS print processor persistence and encrypted spool container
let lookback = 45d;
DeviceRegistryEvents
| where Timestamp > ago(lookback)
| where RegistryKey has @"\Control\Print\Environments\Windows x64\Print Processors\VSPMsg"
    or RegistryValueData =~ "VSPMsg.dll"
| project Timestamp, DeviceName, ActionType, RegistryKey, RegistryValueName, RegistryValueData,
InitiatingProcessFileName, InitiatingProcessCommandLine
| union (
    DeviceFileEvents
    | where Timestamp > ago(lookback)
    | where FolderPath has @"\Windows\System32\spool\prtprocs\x64\VSPMsg.dll"
        or FolderPath has @"\Windows\System32\spool\drivers\color\config.dat"
        or FolderPath has @"\ProgramData\Microsoft Event\PFs\"
    | project Timestamp, DeviceName, ActionType, RegistryKey="", RegistryValueName="",
RegistryValueData="", InitiatingProcessFileName, InitiatingProcessCommandLine
)
| order by Timestamp desc

// Hunt 3: keylogger activation artifacts
let lookback = 45d;
DeviceFileEvents
| where Timestamp > ago(lookback)
| where FolderPath has @"\AppData\Roaming\Microsoft\Vault\lgf.dat"
    or FolderPath has @"\AppData\Roaming\Microsoft\Vault\lgf.dat"
| project Timestamp, DeviceName, ActionType, FolderPath, SHA1, InitiatingProcessFileName,
InitiatingProcessCommandLine

index=edr sourcetype=process earliest=-45d
(FileName="svchost.exe" AND (ParentImage="*\spoolsv.exe" OR ParentImage="*\
ApphostRagistreationVerifier.exe"))
OR CommandLine="*ApphostRagistreationVerifier*"
OR CommandLine="*VSPMsg*"
| stats values(CommandLine) as cmd values(ParentImage) as parent values(User) as user min(_time) as
first max(_time) as last by host FileName
```

14.3 Triage Guidance for Hunt Hits

Meaningful hits usually include more than one artifact class: an abnormal print processor plus suspicious spool files, a scheduled task plus Fonts-directory DLL side-loading, a transient driver load plus suspicious firewall rule, or network IOC traffic plus matching host persistence. Single file-name hits should be enriched with signer, hash, first-seen time, parent process, and prevalence.

Escalate to incident response if any of the following are present: confirmed ESET-published SHA1 on an endpoint; VSPMsg print processor or ApphostRagistreationVerifier scheduled task on a server; RawWNPF, fsdiskbit.sys, bthcam.sys, or msdiskserver artifacts; 207.148.78.36 communications from a Windows server or administrative host; or evidence that local network/process enumeration disagrees with perimeter or EDR telemetry.

15. Triage and Incident Response

Priority	Action	Owner	Rationale
Critical	Isolate confirmed hosts using network controls that do not depend on the compromised OS	IR / Network	Rootkit functionality may hide local processes and sockets
Critical	Acquire volatile memory and preserve disk offline before broad cleanup	IR / Forensics	RawWNPF may hide artifacts during live response
High	Export scheduled tasks, IFEO keys, print processor keys, firewall rules, service-control events, driver-load logs, and spool directory metadata	IR / Endpoint	Preserves persistence and driver evidence
High	Scope C2 traffic to 207.148.78.36 and related Vultr infrastructure, including UDP/53 and WebSocket/80	Network / Detection	C2 may use non-DNS UDP/53 and multi-protocol fallback
High	Rotate credentials exposed on affected hosts, especially service accounts and administrator sessions	Identity	Backdoor supports interactive shell, keylogging, and file access
Medium	Review internet-facing application logs and vulnerability exposure around the first suspicious timestamp	Vulnerability / AppSec	Initial access likely involved public-facing applications but was not confirmed
Medium	Hunt peer servers with similar roles, print services, shared admin accounts, or same subnet	Threat Hunting	FishMonger activity should be scoped as a campaign intrusion
Medium	If firmware or boot anomalies exist, collect firmware state and review CVE-2023-24932 exposure	Platform / IR	ESET reported limited indications of possible UEFI bootkit involvement

Containment guidance: do not trust live host views for processes, files, registry, or network sockets on a suspected WIN_DRV host; prefer network-level isolation, hypervisor snapshots where applicable, trusted boot media, and offline acquisition; preserve spool directories, Fonts directory artifacts, registry hives, System event logs, Code Integrity logs, Sysmon logs, firewall policy, and EDR raw telemetry before removal.

16. Validation and Emulation

Validation Area	Status	Notes
Source verification	Completed	ESET primary report and The Hacker News intake article reviewed on 2026-06-17
Local static reverse engineering	Not performed	No sample was provided to Lost Boys Cyber
Local dynamic execution	Not performed	No sample was provided and no detonation

		was authorized
IOC sidecar creation	Completed	Public ESET IOCs normalized into iocs.csv
Detection content	Completed	Sigma, YARA, KQL, and SPL content authored for analyst validation
ATT&CK mapping	Completed	Based on ESET mapping plus Lost Boys Cyber report structure

Safe emulation options: create benign registry keys under a controlled lab path mirroring Print Processors\VSPMsg and IFEO vds.exe debugger to validate telemetry collection; create harmless test files named ApphostRagistrationVerifier.exe, tpsvcloc.dll, and VSPMsg.dll in a lab directory; simulate scheduled-task creation, print-processor registry creation, and firewall-rule creation; use synthetic PCAP payloads containing documented magic values only in an isolated lab.

Do not emulate RawWNPF driver loading or process doppelganging on production systems. If real samples are acquired later, use an isolated malware-analysis VM with snapshots, kernel debugging controls, and network containment.

17. Recommendations

Priority	Owner	Action	Rationale
Critical	Detection Engineering	Deploy high-severity monitoring for ApphostRagistrationVerifier, VSPMsg, fsdiskbit.sys, bthcam.sys, msdiskserver, RawWNPF, and 207.148.78.36	These are high-confidence public SprySOCKS pivots
Critical	Incident Response	Treat confirmed SprySOCKS artifacts as broader intrusion evidence and scope identities, adjacent servers, and public-facing apps	Backdoor capability supports long-term espionage and follow-on activity
High	Endpoint Engineering	Ensure Sysmon or EDR captures driver loads, image loads, registry writes, scheduled-task changes, firewall changes, and print-processor modifications	Required telemetry for this family is often outside basic process logging
High	Infrastructure	Disable Print Spooler on servers that do not require it and restrict administrative print processor installation	Reduces WIN_PLUS persistence opportunity
High	Vulnerability Management	Review externally exposed Windows servers and applications for unpatched N-day exposure around 2023-2024 and current internet-facing risk	FishMonger initial access commonly targets public-facing servers
High	Network Security	Add C2 IOC and magic-value hunts to IDS/NetFlow review, but avoid brittle IP-only prevention	Infrastructure may change; protocol behavior provides stronger context

Medium	Forensics	Prepare an offline acquisition runbook for suspected rootkit cases	WIN_DRV may blind live host tools
Medium	Identity	Audit privileged logons to affected hosts and rotate credentials after containment	Keylogging, shell access, and file collection increase credential-exposure risk
Medium	Threat Hunting	Run the KQL/SPL hunts across 45-90 days for Windows servers, print servers, and administrative hosts	ESET observed activity across an extended 2023-2024 window

18. References and Refresh Notes

- [1] ESET Research, FishMonger's arsenal upgraded: SprySOCKS for Windows, 2026-06-16, <https://www.welivesecurity.com/en/eset-research/fishmongers-arsenal-upgraded-sprysocks-windows/>
- [2] The Hacker News, China-Linked SprySOCKS Backdoor Expands to Windows with Driver-Based Stealth, 2026-06-16, <https://thehackernews.com/2026/06/china-linked-sprysocks-backdoor-expands.html>
- [3] Trend Micro SprySOCKS Linux reporting referenced by ESET; use as lineage context if original samples or historical Linux SprySOCKS detections are in scope.
- [4] MITRE ATT&CK Enterprise Matrix, version 19 referenced by ESET in the public report.

Refresh notes:

- Re-check ESET and other vendor reporting for SHA256 values before publication if sample hashes are required for client tooling.
- Re-score IP indicators before blocking because Vultr-hosted infrastructure can be reassigned.
- If samples are later obtained, update Sections 5-7 with local static, code, and dynamic evidence and preserve Ghidra/radare2 notes under evidence/.
- If internal telemetry produces confirmed hits, add client-specific process trees, timestamps, user context, and containment outcomes before external distribution.